

Project Demonstration – EduGenie-AI

Google Gemini Powered AI Learning Assistant

Page 1 – Demonstration Overview and Objectives

Project Demonstration is the final stage in which EduGenie-AI is presented as a working application. The demonstration should show the problem, proposed solution, main features, technical workflow, and actual system output.

Demonstration Objectives

- Introduce the purpose of EduGenie-AI.
- Explain how Google Gemini supports learning.
- Show the main web interface.
- Demonstrate each major learning feature.
- Explain the backend and request flow.
- Present testing results and future scope.

Recommended Demonstration Flow

Step	Activity
1	Project introduction and problem statement
2	Objectives and key features
3	Architecture and technology explanation
4	Live application launch
5	Feature-by-feature demonstration
6	Testing and result explanation
7	Future scope and conclusion

Preparation Before Demo

- Start the application and verify the health endpoint.
- Confirm Gemini configuration works.
- Prepare simple test questions and topics.
- Check that all main features are accessible.
- Keep documentation and architecture diagram ready.
- Prepare screenshots as backup in case of connectivity problems.

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Page 2 – Live Application Demonstration

The live demonstration should begin with the EduGenie-AI home page. The presenter can briefly explain the interface before showing individual learning functions.

Home Page Demonstration

- Open the application in a browser.
- Introduce the EduGenie-AI title and purpose.
- Point out the input area and feature controls.
- Explain how a student submits a learning request.
- Show where the generated result appears.

Question Answering Demo

Enter a simple academic question, such as a programming or computer-science concept. Submit the request and show the AI-generated answer. Explain that the backend sends the request to the Gemini model and returns the result to the frontend.

Explanation Demo

Choose an academic topic and request an explanation. Show how EduGenie-AI can provide a structured explanation intended to be easier for a student to understand.

Important Presentation Point

During the live demo, focus on the user experience: input → request → AI processing → educational response. Avoid reading every generated sentence; instead explain what the feature demonstrates.

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Page 3 – Demonstration of Learning Features

The core strength of EduGenie-AI is its collection of learning-oriented features. Each feature should be demonstrated with a short and clear example.

Quiz Demonstration

- Select or enter a topic.
- Request quiz content.
- Show the generated questions.
- Explain how quiz generation can support self-assessment.

Summary Demonstration

- Provide a short learning passage or notes.
- Send the summarization request.
- Display the concise result.
- Explain how summarization can help students revise important points.

Learning Recommendation Demonstration

- Enter a topic the student wants to learn.
- Request learning recommendations.
- Show the suggested sequence or topics.
- Explain how recommendations can organize further study.

Feature Summary

Feature	Demonstration Purpose
Q&A	Get answers to student questions
Explanation	Understand difficult concepts
Quiz	Practice and self-assess
Summary	Revise long learning material
Recommendations	Plan further learning

The presenter should use short, realistic examples so that each feature can be understood quickly.

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Page 4 – Technical Demonstration and Testing Results

After the user-facing demonstration, the technical workflow can be explained briefly. This shows how the frontend, FastAPI backend, learning modules, and Gemini service work together.

Technical Workflow

- Student enters input in the frontend.
- JavaScript sends a request to a FastAPI endpoint.
- FastAPI routes the request to the correct module.
- The module prepares an educational prompt.
- Google Gemini generates the response.
- The backend returns the result.
- JavaScript displays the result to the student.

Testing Demonstration

Area	What to Show
Server	Application starts successfully
Health Check	/health returns a successful status
Gemini	A test request receives an AI response
Features	Major learning functions produce results
Error Handling	Invalid input is handled appropriately
Security	API key remains outside source code

Backup Plan

Because the live AI service depends on network and API availability, keep screenshots or previously captured outputs ready. If a live request fails, explain the intended workflow and use the prepared evidence rather than spending the demonstration time debugging.

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Page 5 – Presentation Script, Future Scope and Conclusion

Suggested Presentation Script

“EduGenie-AI is a Google Gemini powered AI learning assistant designed to help students with question answering, explanations, quizzes, summaries, and learning recommendations. The system uses a web interface connected to a FastAPI backend. The backend communicates with Google Gemini and returns educational responses to the student. The project was developed with modular components so that features can be tested and improved independently.”

Future Scope

- Personalized student accounts and progress tracking.
- Database-based learning history.
- Advanced quiz scoring.
- Voice and multilingual interaction.
- Teacher dashboards and analytics.
- More personalized AI learning paths.

Demonstration Checklist

- Application starts before the presentation.
- Internet connection is checked.
- Gemini configuration is verified.
- Sample inputs are prepared.
- All major features are tested once.
- Screenshots are available as backup.
- Architecture and project documentation are ready.

Conclusion

The Project Demonstration presents EduGenie-AI as a complete academic application, connecting its problem statement, technical architecture, implemented features, and testing results. A clear demonstration helps evaluators understand both the practical use of the system and the development work behind it.

Project: EduGenie-AI | Technology: FastAPI + Google Gemini + HTML/CSS/JavaScript